



(An Empowered Autonomous Institute Affiliated to SPPU, Pune)
NAAC Accredited A+ Grade & NBA
Department of CSE-Artificial Intelligence
Subject :- Advanced Quantum
Computation(23UQCMDL3604)
TAE II-Poster Presentation

AMPLITUDE AMPLIFICATION IN QUANTUM COMPUTING

Introduction :

- Amplitude Amplification is a quantum technique used to increase the probability of correct answers in a quantum system.
- It is a generalization of Grover's Search Algorithm
- It works by increasing the amplitude (probability) of desired states
- Helps in solving problems faster than classical methods

HOW IT WORKS:

Step 1: Initialization (Superposition)

- Start with n qubits initialized to $|0\rangle$
- Apply Hadamard gates to create superposition
- System now represents all possible states equally

Step 2: Oracle Operation (Marking Solution)

- Oracle identifies the correct state
- It flips the phase (sign) of the solution state
- Correct state becomes negative

Step 3: Diffusion Operator (Amplification)

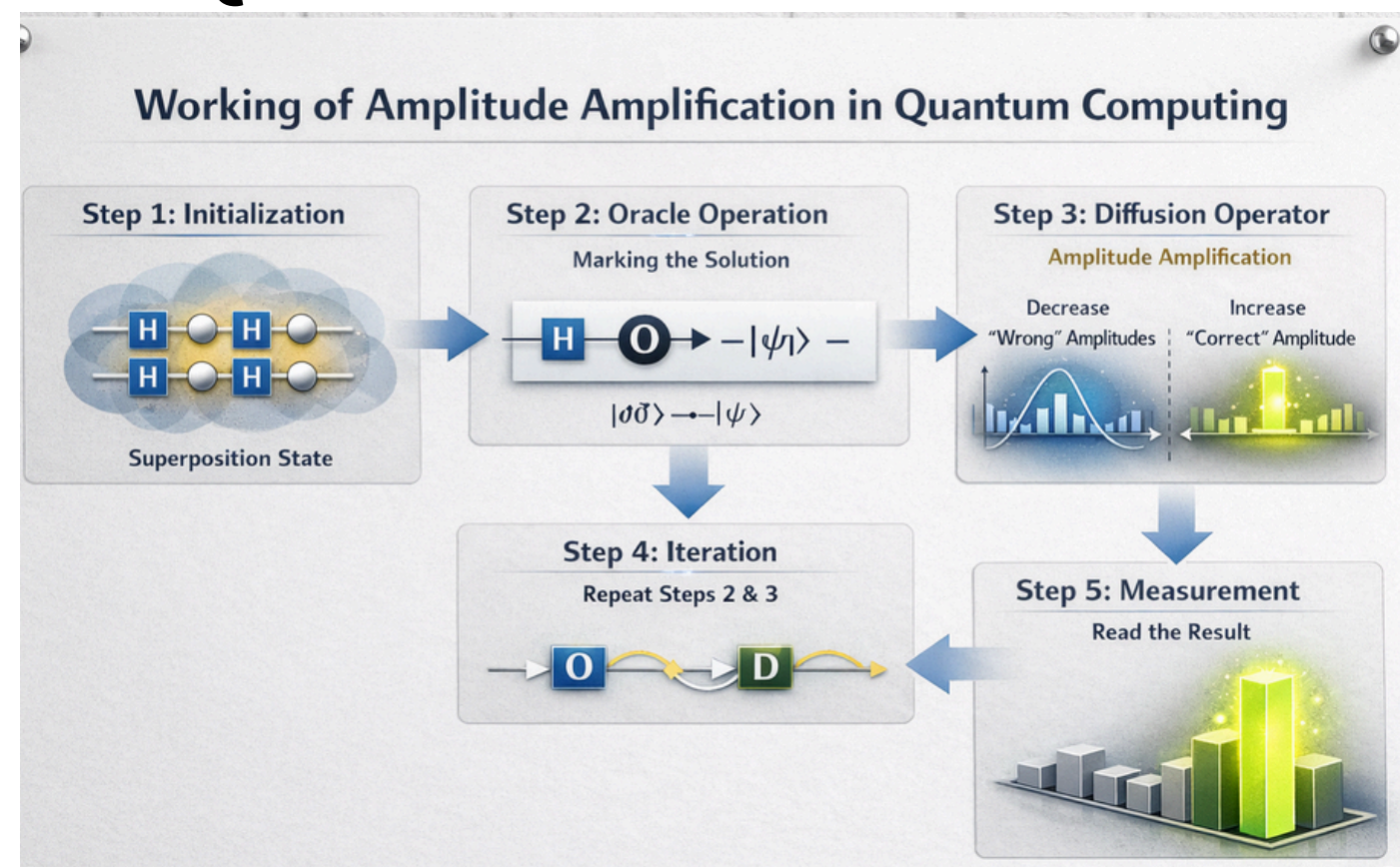
- Also called Inversion About the Mean
- Increases amplitude of marked state
- Decreases amplitude of all other states

Step 4: Iteration Process

- Repeat:
- Oracle $\rightarrow$ Diffusion
- Each iteration:
- Increases probability of correct state

Step 5: Measurement

- After optimal iterations:
- Measure the system
- Highest probability $\rightarrow$ correct solution obtained

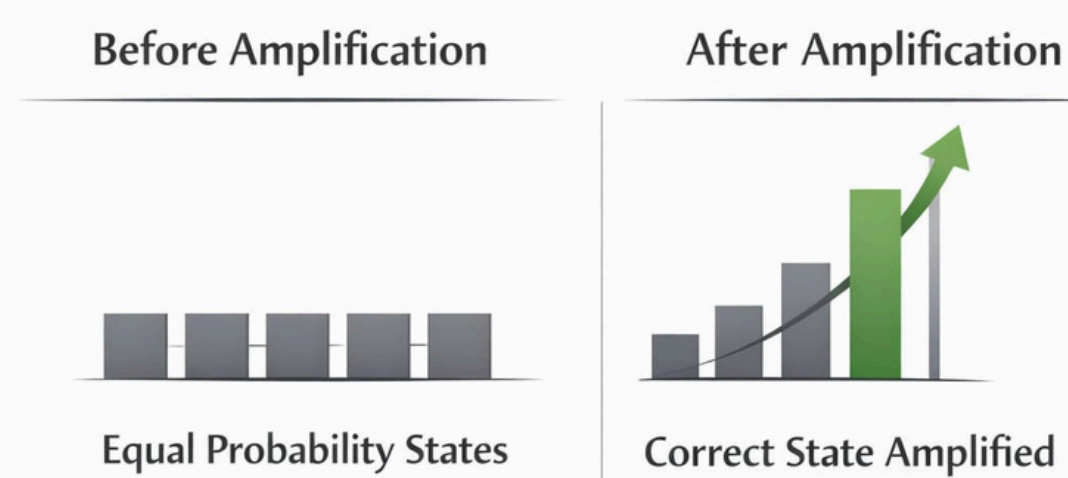


MATHEMATICAL IDEA

- Probability = (Amplitude)²
- After each iteration:
- Correct state amplitude increases
- Wrong state amplitude decreases
- Optimal iterations $\approx \sqrt{N}$

ADVANTAGES

- Faster than classical algorithms
- Reduces search time
- Works for large datasets
- Used in optimization problems



APPLICATIONS:

- Database Search
- Cryptography
- AI & Machine Learning
- Optimization Problems
- Pattern Recognition

Guided by :- Prof. Trupti Mohota

Name: Aarya Gonjari
Class : TY CSE(AI)
Roll no. : A4